



Let's assume that we are given a second order ordinary differential equation.

$$\frac{1}{5} \frac{d^2 x}{dt^2} + 1.2 \frac{dx}{dt} + 2x = 5 \cos 4t$$

$$\Rightarrow \frac{d^2 x}{dt^2} + 6 \frac{dx}{dt} + 10x = 25 \cos 4t$$

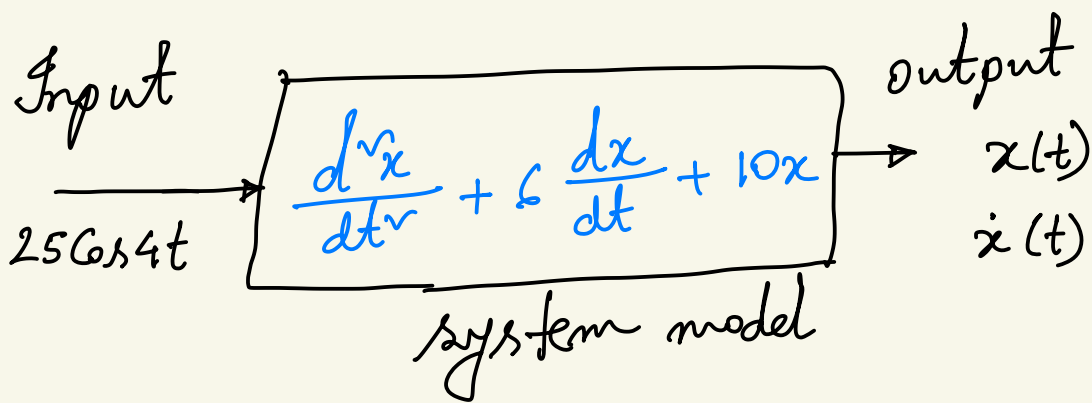
The given initial conditions are

$$x(0) = 0.5$$

$$\frac{dx}{dt}(0) = x'(0) = 0$$

We need to find a solution of this 2nd order differential equation.

This second order differential equation represents a model of a physical system. We need to find a solution of the system, that is, how the system behaves as the time goes on.



From the materials learned in differential equation, you know that

in order to find the complete solution $x(t)$, we need to find the complementary solution $x_c(t)$ and a particular solution $x_p(t)$.

$$x(t) = x_c(t) + x_p(t)$$

For finding complementary solution,
on,

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 10x = 0 \quad \text{--- (1)}$$

$$\text{let, } x = e^{at}$$

$$\Rightarrow \frac{dx}{dt} = a e^{at}$$

$$\Rightarrow \frac{d^2x}{dt^2} = a^2 e^{at}$$

Substitute in eqⁿ ①

$$a^r e^{at} + 6a e^{at} + 10 e^{at} = 0$$

$$\Rightarrow e^{at} (a^r + 6a + 10) = 0$$

$$e^{at} \neq 0 \rightarrow \boxed{a^r + 6a + 10 = 0}$$

characteristic equation

We can find two solutions

$$a_{1,2} = \frac{-6 \pm \sqrt{6^2 - 4 \cdot 1 \cdot 10}}{2 \cdot 1}$$

$$\left. \begin{array}{l} a_1 = -3 + i \\ a_2 = -3 - i \end{array} \right\} \text{Complex number}$$

$$x_c = c_1 e^{a_1 t} + c_2 e^{a_2 t}$$

$$x_c = c_1 e^{(-3+i)t} + c_2 e^{(-3-i)t}$$

$$= c_1 \cdot e^{-3t} \cdot e^{it} + c_2 e^{-3t} \cdot e^{-it}$$

$$= e^{-3t} (c_1 e^{it} + c_2 e^{-it})$$

$$= e^{-3t} \left\{ c_1 (\cos t + i \sin t) + c_2 (\cos t - i \sin t) \right\}$$

$$= e^{-3t} \left\{ c_1 \cos t + c_1 i \sin t + c_2 \cos t - c_2 i \sin t \right\}$$

$$= e^{-3t} \left\{ (c_1 + c_2) \cos t + i (c_1 - c_2) \sin t \right\}$$

$$x_c = e^{-3t} \cdot (A_1 \cos t + B_1 \sin t)$$

In order to find the particular solution, assume

$$x_p = A \cos 4t + B \sin 4t$$

$$\Rightarrow \frac{dx_p}{dt} = -4A \sin 4t + 4B \cos 4t$$

$$\Rightarrow \frac{d^2 x_p}{dt^2} = -16A \cos 4t - 16B \sin 4t$$

Substitute in the original equation :

$$\frac{d^2 x_p}{dt^2} + 6 \frac{dx_p}{dt} + 10 x_p = 25 \cos 4t$$

$$\Rightarrow (-16A \cos 4t - 16B \sin 4t)$$

$$+ (-24A \sin 4t + 24B \cos 4t)$$

$$+ (10A \cos 4t + 10B \sin 4t)$$

$$= 25 \cos 4t$$

Group similar terms:

$$\cos 4t (-16A + 24B + 10A) +$$

$$\sin 4t (10B - 16B - 24A)$$

$$= 25 \cos 4t$$

$$\Rightarrow \cos 4t (24B - 6A) + \sin 4t (-6B - 24A)$$

$$= 25 \cos 4t$$

we get,

$$\left. \begin{array}{l} 24B - 6A = 25 \\ -6B - 24A = 0 \end{array} \right\}$$

$$\text{solving} \rightarrow B = \frac{50}{51}$$

$$A = -\frac{25}{102}$$

$$x_p = -\frac{25}{102} \cos 4t + \frac{50}{51} \sin 4t$$

Total Solution

$$x(t) = e^{-3t} \left(A_1 \cos t + B_1 \sin t \right) \\ - \frac{25}{102} \cos 4t + \frac{50}{51} \sin 4t$$

We still need to find A_1 and B_1 from initial conditions.

$$\text{when } t=0, x(0) = 0.5$$

$$x'(0) = 0$$

$$A_1 = \frac{38}{51}$$

$$B_1 = -\frac{86}{51}$$

Complete solution

$$x(t) = e^{-3t} \left(\frac{38}{51} \cos t - \frac{86}{51} \sin t \right) \\ - \frac{25}{102} \cos 4t + \frac{50}{51} \sin 4t$$

(Ans.)
This is the analytical solution.

